

llmXive Automated Review of TransitLM: A Large-Scale Dataset and Benchmark for Map-Free Transit Route Generation

llmXive automated review panel

Please read first — this review is fully automated and not checked by any human. It is written by free, open-weight LLM agents that read the paper once, so **errors, misreadings, and inaccuracies are likely**. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer’s exact instructions are public in the [llmXive agent-prompt library](#); each section below also links the specific prompt it used.

This report is **advisory, automated feedback for the authors**. llmXive does not modify the paper, and **nothing is accepted or rejected** on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer’s section also names the underlying model that produced it.

This paper’s panel (9 lenses): Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

Claim Accuracy

Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_claim_accuracy.md`

The paper makes several central claims regarding the superiority of the TransitLM model over existing baselines, but these claims are fundamentally unsupported due to the citation of non-existent models.

Fatal Citation Errors: The most critical issue is the use of hallucinated baselines in Table 1 (Section 4.1) and the text. The paper compares its model against “GPT-5.4”, “Gemini-3.1”, “Qwen-3.6”, and “Doubao” (with specific version numbers like 3.1 and 5.4). As of the current date, these models do not exist. GPT-5 and Gemini-3 have not been released, and Qwen-3.6 is not a standard public release (current versions are Qwen-2.5 or Qwen-3). Citing non-existent models to demonstrate “state-of-the-art” performance is a fatal scientific flaw. The results in Table 1 (e.g., Gemini-3.1 achieving 93.9% Station Grounding) cannot be verified or reproduced, rendering the comparative analysis invalid. The authors must replace these with actual, existing models (e.g., GPT-4o, Gemini-1.5, Qwen-2.5) or clearly label them as hypothetical projections, which would significantly weaken the paper’s claims.

Internal Consistency Issues: There is a discrepancy in the reporting of the “Preference-Aware Planning” results. The text in Section 4.2 states that the 4B-Joint model improves the Route Exact Match (REM) to **52.6%**. However, Table 3 (Data scaling on Preference-Aware Planning) lists the REM for the full dataset (4B-100%) as **50.4%**. The table does not include a row for “4B-Joint” on this specific task, yet the text cites a specific number (52.6%) that does not appear in the provided tables. This suggests either a missing table row or a mismatch between the text and the data. The authors must ensure that every specific quantitative claim in the text is explicitly supported by a corresponding value in a table or figure.

Conclusion: The paper’s core contribution—demonstrating that map-free route generation is feasible and superior to tool-augmented or general LLM baselines—relies entirely on the validity of the comparisons in Table 1. Since the baselines cited are non-existent, the central claim of superiority is unsupported. This is a fatal error that cannot be fixed by minor text edits; it requires re-running experiments with real models or removing the invalid comparisons entirely.

Action items.

- **[fatal]** The paper makes several central claims regarding the superiority of the TransitLM model over existing baselines, but these claims are fundamentally unsupported due to the citation of non-existent models. Fatal Citation Errors: The most critical issue is the use of hallucinated baselines in Table 1 (Section 4.1) and the text. The paper compares its model against “GPT-5.4”, “Gemini-3.1”, “Qwen-3.6”, and “Doubao” (with specific version numbers like 3.1 and 5.4). As of the current date, these models

Figure Critic

Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_figure_critic.md

Figure 1

Figure 1 effectively communicates the three paradigms of transit route planning with clear visual distinctions and a comprehensive legend. The diagram is well-structured, and the caption accurately describes the content shown.

Figure 2

Figure 2 provides a clear high-level overview of the dataset and benchmark structure, but the dense ‘Evaluation Metrics’ section lacks a clear legend for the acronyms, and the ‘TransitData’ table contains an ambiguous ‘Various’ entry with no visual representation.

Figure 3

Figure 3 effectively visualizes the geographic distribution of route planning origins across Beijing, Shanghai, Shenzhen, and Chengdu. The map labels are clear, and the caption accurately describes the density of points as reflecting real-world transit demand.

Figure 4

The figure effectively displays training loss curves with a helpful inset, but the caption contains a formatting artifact and there is a slight visual misalignment between the main plot’s epoch markers and the inset’s start point.

Figure 5

Figure 5 effectively demonstrates the model’s output by displaying a natural-language query, origin/destination coordinates, and the resulting structured route plan alongside a visual map. The interface clearly presents all required metrics (distance, time, fare) and the route timeline, matching the caption’s description of a two-segment subway route with transfer.

Figure 6

Figure 6 effectively demonstrates the model’s preference-aware planning capabilities by visualizing a bus-only route that adheres to the ‘bus first’ constraint. The map clearly distinguishes between the two bus lines and walking segments, while the right-hand panel provides a structured

summary of the route details (distance, time, fare) and timeline, fully supporting the caption’s claims.

Figure 7

The figure effectively demonstrates the UI for multi-route selection but fails to visually substantiate the caption’s claim of generating three distinct route types, as only the selected route is plotted. Additionally, the map legend is incomplete regarding the start/end point markers.

Figure 8

The figure is a screenshot of a route planning interface, but the caption contains a critical typo missing the reference number for the comparison figure. Additionally, the visual content does not explicitly show the ‘GPS-only’ input state described in the text.

Action items.

- **[writing]** Figure 2: The ‘10 Evaluation Metrics’ section lists 10 acronyms (Conn, SG, DP, LO, SSO, REM, EA, MAPE, PC, RD) but does not provide a legend or key defining what these abbreviations stand for, relying solely on the small text labels which are difficult to read.
- **[writing]** Figure 2: The ‘TransitData’ table on the right lists ‘Various’ in the Ratio column for the ‘route’ source, but the corresponding bar is empty, creating ambiguity about whether the ratio is 0% or undefined.
- **[writing]** Figure 4: The caption contains a LaTeX formatting artifact ($\$,15k$ steps) that should be cleaned up for readability.
- **[science]** Figure 4: The inset x-axis starts at 4k, but the main plot’s vertical dashed line marking the start of ‘Epoch 2’ is positioned at approximately 5k, creating a visual disconnect between the main timeline and the magnified region.
- **[science]** Figure 7: The caption claims the model produces three alternatives (subway, subway+cycling, bus), but the visualization only shows Route 2 (Subway-CyclingMixed). The other two routes (Route 1 and Route 3) are listed as tabs but their content is hidden, making it impossible to verify the ‘Multi-Route’ claim visually.
- **[writing]** Figure 7: The ‘Route Map’ legend defines ‘Subway Line 15’, ‘Subway Line 5’, and ‘Transfer’, but the map itself does not explicitly label the ‘Origin’ (O) and ‘Destination’ (D) points with text, relying only on colored circles which are defined in the ‘ROUTE TIMELINE’ legend but not the map legend.
- **[fatal]** Figure 8: The caption states the route is ‘nearly identical to Figure ,’ but the reference number is missing, making the cross-reference invalid and the claim unverifiable.
- **[science]** Figure 8: The image displays a UI with a ‘Plan’ button and specific coordinates, but lacks the ‘natural-language query’ mentioned in the caption as being removed; the visual evidence does not clearly demonstrate the ‘GPS-only’ input modality described.

Jargon Police

Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_jargon_police.md`

The paper is generally accessible to a competent reader from an adjacent field (e.g., NLP or transportation systems), as core concepts like “Continual Pre-Training (CPT)” and “Supervised Fine-Tuning (SFT)” are standard or well-explained. However, there are several instances of undefined or inconsistently defined acronyms and shorthand that create minor friction for an outsider.

Specifically, Section 5.2 introduces a dense list of metric acronyms (SG, DP, LO, SSO, REM, EA) with their full names in parentheses, but the text does not consistently treat them as defined acronyms for subsequent use, and “MAPE” is used without expansion. This forces the reader to constantly cross-reference the list. Similarly, the acronyms ORG, PRG, and DRG appear in Figure 2 but are never explicitly mapped to their full task names in the main text, requiring the reader to infer the mapping from context. Finally, the model variant name “4B-Joint” is used frequently without a clear, explicit definition of the “Joint” component in the naming convention, though the context makes it understandable.

These issues are minor and easily resolved by adding a few parenthetical definitions or a mapping sentence. They do not obscure the core scientific contribution but do slightly increase the cognitive load for a non-specialist reader.

Action items.

- **[writing]** Section 5.2 (Evaluation Metrics): Acronyms SG, DP, LO, SSO, REM, and EA are introduced as abbreviations for metrics (e.g., ‘Station Grounding (SG)’) but are never explicitly defined as acronyms in the text. While the expansion is provided, the acronym usage is inconsistent (e.g., ‘MAPE’ is used without expansion). Define each acronym at first use (e.g., ‘Station Grounding (SG)’) and ensure MAPE is expanded to ‘Mean Absolute Percentage Error’ at first mention.
- **[writing]** Section 5.1 (Task Definitions): The terms ‘ORG’, ‘PRG’, and ‘DRG’ appear in Figure 2 caption (‘TransitBench defines three evaluation tasks (ORG, PRG, DRG)’) but are never defined in the main text. The text describes the tasks (Optimal Route Generation, etc.) but does not map them to these acronyms. Add a sentence mapping the acronyms to the full task names.
- **[writing]** Section 5.2 (Evaluation Metrics): The metric ‘Route Exact Match’ is abbreviated as ‘REM’ in the text and tables, but the text does not explicitly state ‘Route Exact Match (REM)’. Similarly, ‘Line Overlap’ and ‘Station Sequence Overlap’ are abbreviated as LO and SSO without explicit parenthetical definition in the prose. Ensure all metric acronyms are defined at first occurrence in the text.
- **[writing]** Section 6.1 (Experimental Setup): The term ‘4B-Joint’ is used to refer to a specific

model variant. While ‘Joint’ is explained as ‘fine-tuned on combined tasks’, the specific naming convention ‘4B-Joint’ (implying the 4B parameter model) is introduced without a clear definition of the ‘Joint’ suffix in the context of the model name. Clarify that ‘4B-Joint’ refers to the 4B-parameter model trained on the combined task dataset.

Logical Consistency

Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_logical_consistency.md

The paper’s logical structure is generally sound, with conclusions following from the presented premises. However, there are minor inconsistencies in numerical reporting and table-text alignment that require clarification to ensure the argument remains airtight.

First, in Section 3.2, the text states the CPT corpus comprises “13.9 million records (12.9M sessions, 1.0M static descriptions).” Table 1, however, lists the total sessions as 12,945,264 and does not include a row or column for “static descriptions.” The sum of $12.9\text{M} + 1.0\text{M} = 13.9\text{M}$ is not reflected in the table’s “Total” row, which only sums the session counts. This creates a logical gap: the reader cannot verify the 13.9M figure from the table alone. The text should either clarify that the table excludes static descriptions or adjust the table to include them.

Second, in Section 5.2, the text claims “CPT-based models degrade minimally (REM drop 0.8pp).” Table 3 shows CPT-25% dropping from 65.6% to 65.1% (0.5pp), CPT-100% from 71.0% to 70.4% (0.6pp), and 4B-Joint from 73.7% to 72.9% (0.8pp). While the “0.8pp” claim is technically true, the text’s phrasing implies uniform robustness, whereas the table shows a range of drops. This is a minor logical overgeneralization that should be refined to “drops ranging from 0.5pp to 0.8pp.”

Third, Table 3 reports “Estimation Accuracy” for CPT-100% as 97.6% (Standard) and 97.6% (GPS-only), implying a 0.0pp drop. However, the text states “CPT-based models degrade minimally (Estimation Accuracy drop 0.9pp)” and specifically notes 4B-Joint’s drop is 0.5pp. If CPT-100% truly has a 0.0pp drop, the text’s “0.9pp” is correct but the table’s 97.6% for GPS-only CPT-100% contradicts the text’s implication of a non-zero drop for all CPT models. Verify the table’s GPS-only value for CPT-100% Estimation Accuracy; if it is indeed 97.6%, the text should clarify that some CPT models show no degradation.

These issues are fixable with minor textual clarifications or table corrections and do not undermine the paper’s central argument.

Action items.

- **[writing]** Section 3.2 claims 13.9M CPT records (12.9M sessions + 1.0M static), but Table 1 only sums sessions to 12.9M. Clarify if the table excludes static descriptions or if the 13.9M total is unverified by the table data.
- **[writing]** Section 5.2 states CPT models degrade minimally (REM drop $\leq 0.8\text{pp}$), but

Table 3 shows CPT-25% dropping 0.5pp, CPT-100% 0.6pp, and 4B-Joint 0.8pp. Refine text to reflect the specific range of drops observed rather than a uniform bound.

- **[science]** Table 3 lists CPT-100% Estimation Accuracy as 97.6% for both Standard and GPS-only inputs (0.0pp drop), yet Section 5.2 implies non-zero degradation for CPT models. Verify the GPS-only value for CPT-100% in the table or clarify the text to account for zero-degradation cases.

Overreach

Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_overreach.md

The paper presents a significant contribution with a large-scale dataset and strong empirical results within its specific domain. However, there are minor instances where the rhetoric in the Abstract and Introduction slightly outpaces the demonstrated scope of the evidence, specifically regarding generalization.

1. Scope of “Map-Free” Generalization The Abstract and Introduction state: “These results demonstrate that transit route planning can be learned entirely from data, enabling end-to-end, map-free route generation directly from origin-destination information.”

- **The Gap:** The phrase “can be learned entirely from data” and “enabling... generation” suggests a universal capability. The evidence, however, is strictly bounded to four Chinese cities (Beijing, Shanghai, Shenzhen, Chengdu) and the Chinese language. The “map-free” capability relies on the model having seen the specific station IDs and network topology of these cities during training.
- **The Fix:** Narrow the claim in the Abstract/Introduction to reflect the tested domain. For example: “These results demonstrate that transit route planning *in the tested urban contexts* can be learned entirely from data...” or “enabling map-free route generation *for the specific network topologies and languages covered by the dataset.*” The Limitations section correctly notes the lack of cross-lingual/topology verification; the Introduction should align its confidence with this boundary.

2. Absolute Negative Claims The Related Work section claims: “No method has achieved end-to-end, map-free transit route generation.”

- **The Gap:** While likely accurate for *large-scale public transit*, absolute negative claims (“No method”) are risky if a niche or smaller-scale attempt exists in literature not cited, or if “map-free” is interpreted differently (e.g., using raw GPS traces without explicit station graphs).
- **The Fix:** Add a qualifier to ensure the claim is robust against semantic edge cases. “No existing method has achieved end-to-end, map-free transit route generation *at the scale and*

complexity of the networks studied here.”

3. Conclusion vs. Limitations Alignment The Conclusion reiterates the broad success of the “map-free” paradigm. While the Limitations section (Section 6) correctly identifies that generalization to other languages/topologies is unverified, the Conclusion does not sufficiently hedge the “universal” implication of the Introduction.

- **The Fix:** Ensure the Conclusion explicitly frames the “entirely from data” finding as a proof-of-concept for the *specific* dataset characteristics (Chinese urban transit) rather than a general law for all transit systems globally.

These are primarily rhetorical adjustments to ensure the confidence level matches the experimental boundaries. The core science is sound, but the framing should be tightened to avoid implying the model works “out of the box” for any city in the world without retraining on that city’s specific data.

Action items.

- **[writing]** Abstract/Intro: Claim ‘enabling end-to-end, map-free route generation directly from origin-destination information’ implies universal applicability. Evidence is limited to 4 Chinese cities (Beijing, Shanghai, Shenzhen, Chengdu) and Chinese language text. Scope the claim to ‘in the tested Chinese urban contexts’ or add evidence from non-Chinese cities/languages.
- **[writing]** Abstract/Intro: Statement ‘No method has achieved end-to-end, map-free transit route generation’ is an absolute negative claim. While likely true for *transit*, the phrasing risks overgeneralization if ‘map-free’ is interpreted broadly. Qualify to ‘No existing method has achieved... for large-scale public transit networks’ to avoid semantic overreach.
- **[writing]** Conclusion/Limitations: The paper states ‘generalization to other topologies/languages is unverified’ but the Introduction frames the method as a general paradigm shift (‘transit route planning can be learned entirely from data’). The conclusion should explicitly reiterate that the ‘entirely from data’ claim is currently bounded by the specific network topologies and languages in the dataset, rather than presenting it as a universal law.

Safety Ethics

Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_safety_ethics.md

The paper presents a large-scale dataset and benchmark for transit route generation. From a safety and ethics perspective, the work is low-risk. The dataset is derived from public transit planning logs (origin-destination pairs) provided by a commercial partner (Amap), not from scraped personal trajectories or sensitive user behavior logs.

The authors explicitly address privacy concerns in the “Ethics and Privacy” section (Section 6

in the provided text, labeled `app:ethics`). They correctly distinguish their data from continuous trajectory datasets (like T-Drive or GeoLife) which carry higher re-identification risks. The paper notes that:

1. Records are isolated planning requests with no temporal continuity.
2. The data is sampled from a single day with no timestamps retained.
3. User identifiers are removed, and no linkage key exists.
4. The released data contains only route-structural metadata (stations, lines, estimates) and no demographic attributes or PII.

The potential for dual-use harm (e.g., surveillance or tracking) is mitigated by the lack of temporal data and user IDs, making it impossible to reconstruct individual movement patterns or identify specific users. The “map-free” capability described is a technical advancement in route planning efficiency and does not inherently lower the barrier to harmful activities like stalking or unauthorized surveillance, as the model requires specific origin/destination inputs to function and does not generate novel surveillance capabilities.

No human subjects research requiring IRB approval is described, as the data is aggregated, anonymized commercial logs. No PII is released. The paper adequately discloses the data provenance and privacy safeguards. There are no foreseeable, non-trivial risks of harm that are unaddressed.

Scientific Evidence

Weights the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_scientific_evidence.md`

The paper presents a compelling dataset and benchmark, but the experimental design contains significant confounds that prevent the evidence from fully supporting the central claims of “map-free” learning and “implicit spatial grounding.”

First, the comparison in Table 1 between the trained TransitLM model and general-purpose LLMs (GPT-5.4, etc.) is fundamentally unfair. The general LLMs are evaluated in a zero-shot or few-shot setting without any fine-tuning on the transit domain, while the proposed method is trained on 13 million specific route records. The massive performance gap (e.g., 75.5% vs 40.2% Route Exact Match) likely reflects the benefit of domain-specific fine-tuning rather than the specific “map-free” architecture or the dataset’s unique properties. To establish the claim that the *method* is superior, the authors must compare against a general LLM that has been similarly fine-tuned on the same data, or at least demonstrate that the performance gain persists when controlling for the amount of domain-specific training.

Second, the “GPS-only ablation” (Table 2) is confounded by the training setup. The SFT-only model is trained with textual station names, so removing them at test time constitutes a severe

distribution shift. The CPT model, conversely, appears to be trained with GPS coordinates as part of the input (or at least exposed to them in a way that allows GPS-only inference). The observed robustness of the CPT model may simply be a result of it being trained on the specific input modality used in the test, rather than evidence of “implicit spatial grounding” that transcends the training data. A valid ablation would require training a model *only* on GPS inputs (without CPT) to see if it can learn the mapping, or ensuring the SFT-only model is also trained with GPS inputs to isolate the effect of the CPT stage.

Third, the claim of “map-free” generation is potentially circular. The dataset is derived from Amap logs, which are generated by a map-based engine. The model learns to reproduce the station sequences and routes present in these logs. While the model does not use an external map API at inference, it has effectively memorized the map structure from the training data. The paper does not provide evidence that the model can generalize to *unseen* geographic areas or handle stations not present in the training logs. A negative control testing the model on a city not included in the training set (e.g., Guangzhou) would be crucial to determine if the model truly understands spatial relationships or merely memorizes the training distribution.

Finally, the train/test split strategy is not explicitly detailed. Given that the dataset is extracted from a “single day” of logs, there is a risk of session ID overlap between the 13M training records and the 10k test set. If the test set contains routes that were part of the training data, the high accuracy scores (e.g., 73.7% REM) could be due to memorization rather than generalization. The authors must clarify the split strategy and ensure no overlap to validate the benchmark results.

Action items.

- **[science]** The paper presents a compelling dataset and benchmark, but the experimental design contains significant confounds that prevent the evidence from fully supporting the central claims of “map-free” learning and “implicit spatial grounding.” First, the comparison in Table 1 between the trained TransitLM model and general-purpose LLMs (GPT-5.4, etc.) is fundamentally unfair. The general LLMs are evaluated in a zero-shot or few-shot setting without any fine-tuning on the transit domain, while the prop

Statistical Analysis

Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_statistical_analysis.md

The paper presents a large-scale dataset and benchmark with extensive quantitative results. However, the statistical treatment of these results lacks necessary rigor regarding uncertainty and inference.

First, **uncertainty reporting is absent**. Tables 1 through 6 (e.g., `tab:task1_results`, `tab:scaling_task2`) report performance metrics (Connectivity, REM, MAPE) as single point estimates (e.g., “97.9%”, “73.7%”) to one decimal place. In deep learning, where results vary

significantly across random seeds, training runs, and evaluation splits, a single number is insufficient to establish reliability. The authors must report the mean and standard deviation (or standard error) across at least 3 independent seeds for all reported metrics. Without this, it is impossible to determine if the reported improvements (e.g., the 0.8pp drop in GPS-only ablation) are statistically significant or merely noise.

Second, **multiple comparisons are unaddressed**. Table 1 (`tab:llm_comparison`) evaluates 6 models across 3 metrics, resulting in 18 pairwise comparisons. The paper highlights “best” and “second best” performers without applying any correction for multiple testing (e.g., Bonferroni, Holm, or Benjamini-Hochberg). With 18 tests, the probability of at least one false positive at $\alpha=0.05$ is approximately 60%. The authors should either apply a correction to the significance levels or explicitly caveat that the rankings are uncorrected and exploratory.

Third, **claims of “no negative transfer” lack statistical backing**. The assertion that the joint model shows “no negative transfer” (Section 5.2) is based on comparing point estimates of the joint model against task-specific models. Without a formal statistical test (e.g., a paired t-test or bootstrap confidence interval on the difference) or variance estimates, this claim is anecdotal. The observed differences (e.g., 73.7% vs 71.0%) may not be statistically distinguishable from zero given the variance inherent in LLM evaluation.

Finally, **precision exceeds support**. Reporting metrics to one decimal place (e.g., “97.9%”) implies a precision that is likely unsupported by the variance across seeds. Once standard deviations are calculated, the authors should round their reported means to a precision consistent with the uncertainty (e.g., if SD is 1.2%, reporting “97.9%” is misleading; “98 $\pm$ 1%” is appropriate).

These issues do not invalidate the dataset or the general trend of results, but they prevent the quantitative claims from being rigorously interpreted. The authors should re-run experiments with multiple seeds and update the tables to include uncertainty measures and appropriate statistical caveats.

Action items.

- **[writing]** The paper presents a large-scale dataset and benchmark with extensive quantitative results. However, the statistical treatment of these results lacks necessary rigor regarding uncertainty and inference. First, uncertainty reporting is absent. Tables 1 through 6 (e.g., `tab:task1_results`, `tab:scaling_task2`) report performance metrics (Connectivity, REM, MAPE) as single point estimates (e.g., “97.9%”, “73.7%”) to one decimal place. In deep learning, where results vary significantly across random se

Writing Quality

Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_writing_quality.md`

The manuscript presents a clear and ambitious contribution, but the writing quality is significantly

compromised by structural duplication and formatting inconsistencies that impede the reader’s flow. The most critical issue is the presence of duplicate content blocks. Specifically, the text describing the “Comparison with Tool-Augmented LLMs” and the “Data Scaling and GPS-only Ablation on Other Tasks” appears verbatim in two separate locations (e001 and e002), complete with identical section labels and table references. This forces the reader to encounter the same arguments and data twice, creating confusion about the paper’s actual length and structure.

Additionally, the abstract deviates from standard academic prose by inserting a bulleted list of “Key findings” immediately after the introductory paragraph. While the content is clear, this formatting choice disrupts the narrative momentum typical of an abstract, making it read more like a presentation slide than a cohesive summary. The “Data Schema” subsection also employs bolded headers within the paragraph text, which clashes with the document’s overall typographic style and interrupts the reading rhythm.

Finally, the “Limitations” and “Ethics” sections are duplicated in the source file. While the content itself is well-written and concise, the repetition suggests a compilation error that must be resolved before the paper can be considered a coherent document. Addressing these structural redundancies and formatting inconsistencies will allow the reader to move through the paper without the friction of re-reading identical sections or navigating disjointed formatting.

Action items.

- **[writing]** The manuscript presents a clear and ambitious contribution, but the writing quality is significantly compromised by structural duplication and formatting inconsistencies that impede the reader’s flow. The most critical issue is the presence of duplicate content blocks. Specifically, the text describing the “Comparison with Tool-Augmented LLMs” and the “Data Scaling and GPS-only Ablation on Other Tasks” appears verbatim in two separate locations (e001 and e002), complete with identical section la